

Chapter 9: Percentage

Comprehensive Solutions for Exercise 9B (Exercise 9.2)

Class VII / VIII Mathematics Study Guide

Key Concepts & Formulas

Before diving into the solutions, let's recall the core mathematical principles of percentages used in this exercise:

1. **Definition of Percentage:** The term 'per cent' means 'per hundred' or 'out of 100'. Thus, a percentage $x\%$ represents a fraction with a denominator of 100:

$$x\% = \frac{x}{100}$$

2. **Finding a Percentage of a Given Quantity:** To find $y\%$ of a quantity Q , convert the percentage into a fraction and multiply it by the quantity:

$$y\% \text{ of } Q = \frac{y}{100} \times Q$$

3. **Expressing One Quantity as a Percentage of Another:** To express quantity A as a percentage of quantity B , write it as a fraction and multiply by 100. Both quantities must be in the **same units**:

$$\text{Required Percentage} = \left(\frac{A}{B} \times 100 \right) \%$$

4. **Finding the Original Number when a Percentage is Given:** If $p\%$ of an unknown number x is k , set up the equation:

$$\frac{p}{100} \times x = k \implies x = \frac{k \times 100}{p}$$

Exercise 9B Step-by-Step Solutions

Question 1

If 16% of a number is 60, find the number.

Solution:

Let the required number be x .

According to the question:

$$16\% \text{ of } x = 60$$

$$\frac{16}{100} \times x = 60$$

$$x = 60 \times \frac{100}{16}$$

$$x = 15 \times 25$$

$$x = 375$$

Therefore, the required number is **375**.

Question 2

If $13\frac{1}{3}\%$ of a number is 90, find the number.

Solution:

Let the required number be x .

First, convert the mixed fraction percentage into an improper fraction:

$$13\frac{1}{3}\% = \frac{13 \times 3 + 1}{3}\% = \frac{40}{3}\%$$

According to the question:

$$\frac{40}{3}\% \text{ of } x = 90$$

$$\left(\frac{40}{3 \times 100}\right) \times x = 90$$

$$\frac{40}{300} \times x = 90$$

$$\frac{2}{15} \times x = 90$$

$$x = 90 \times \frac{15}{2}$$

$$x = 45 \times 15$$

$$x = 675$$

Therefore, the required number is **675**.

Question 3

If 0.6% of a number is 15, find the number.

Solution:

Let the required number be x .

According to the question:

$$0.6\% \text{ of } x = 15$$

$$\frac{0.6}{100} \times x = 15$$

$$x = \frac{15 \times 100}{0.6}$$

$$x = \frac{15000}{6}$$

$$x = 2500$$

Therefore, the required number is **2500**.

Question 4

If $\frac{3}{4}\%$ of a number is 9, find the number.

Solution:

Let the required number be x .

According to the question:

$$\begin{aligned}\frac{3}{4}\% \text{ of } x &= 9 \\ \frac{3}{4 \times 100} \times x &= 9 \\ \frac{3}{400} \times x &= 9 \\ x &= 9 \times \frac{400}{3} \\ x &= 3 \times 400 \\ x &= 1200\end{aligned}$$

Therefore, the required number is **1200**.

Question 5

There are 42 boys and 18 girls in a class. What is the percentage of boys in the class?

Solution:

Given:

Number of boys = 42

Number of girls = 18

Total number of pupils in the class = Number of boys + Number of girls:

$$\text{Total pupils} = 42 + 18 = 60$$

The percentage of boys in the class is:

$$\begin{aligned}\text{Percentage of boys} &= \left(\frac{\text{Number of boys}}{\text{Total pupils}} \times 100 \right) \% \\ &= \left(\frac{42}{60} \times 100 \right) \% \\ &= \left(\frac{7}{10} \times 100 \right) \% \\ &= 70\%\end{aligned}$$

Therefore, the percentage of boys in the class is **70%**.

Question 6

A team won 6 hockey matches and lost 9 matches. What per cent of the matches did the team win?

Solution:

Given:

$$\text{Number of matches won} = 6$$

$$\text{Number of matches lost} = 9$$

Total matches played = Matches won + Matches lost:

$$\text{Total matches} = 6 + 9 = 15$$

The percentage of matches won by the team is:

$$\begin{aligned}\text{Percentage of matches won} &= \left(\frac{\text{Matches won}}{\text{Total matches}} \times 100 \right) \% \\ &= \left(\frac{6}{15} \times 100 \right) \% \\ &= \left(\frac{2}{5} \times 100 \right) \% \\ &= 40\%\end{aligned}$$

Therefore, the team won **40%** of their matches.

Question 7

A batsman scored 75 runs which included 3 boundaries and 8 sixes. What per cent of his total score did he make by running between the wickets?

Solution:

Given:

$$\text{Total runs scored} = 75$$

$$\text{Number of boundaries (4s)} = 3$$

$$\text{Number of sixes (6s)} = 8$$

Let us find the total runs scored through boundaries and sixes:

$$\text{Runs from boundaries} = 3 \times 4 = 12 \text{ runs}$$

$$\text{Runs from sixes} = 8 \times 6 = 48 \text{ runs}$$

$$\text{Total runs scored from boundaries and sixes} = 12 + 48 = 60 \text{ runs}$$

The runs scored by running between the wickets:

$$\text{Runs by running} = \text{Total runs} - \text{Boundary runs} = 75 - 60 = 15 \text{ runs}$$

The percentage of runs made by running is:

$$\begin{aligned} \text{Percentage of runs by running} &= \left(\frac{\text{Runs by running}}{\text{Total runs}} \times 100 \right) \% \\ &= \left(\frac{15}{75} \times 100 \right) \% \\ &= \left(\frac{1}{5} \times 100 \right) \% \\ &= 20\% \end{aligned}$$

Therefore, the batsman made **20%** of his total score by running between the wickets.

Question 8

12% of a sum of money is Rs 42. What is 20% of the same sum?

Solution:

Let the total sum of money be Rs S .

According to the question:

$$12\% \text{ of } S = 42$$

$$\frac{12}{100} \times S = 42$$

$$S = \frac{42 \times 100}{12}$$

$$S = \frac{7 \times 100}{2} = 350$$

So, the total sum of money is Rs **350**.

Now, we need to find 20% of this same sum:

$$\begin{aligned} 20\% \text{ of } S &= 20\% \text{ of } 350 \\ &= \frac{20}{100} \times 350 \\ &= \frac{1}{5} \times 350 = 70 \end{aligned}$$

Therefore, 20% of the same sum is **Rs 70**.

Question 9

$6\frac{1}{4}\%$ of a weight is 0.25 kg. What is 45% of this weight?

Solution:

Let the total weight be W kg.

First, convert the mixed fraction percentage into an improper fraction:

$$6\frac{1}{4}\% = \frac{6 \times 4 + 1}{4}\% = \frac{25}{4}\%$$

According to the question:

$$\begin{aligned}\frac{25}{4}\% \text{ of } W &= 0.25 \\ \left(\frac{25}{4 \times 100}\right) \times W &= 0.25 \\ \frac{25}{400} \times W &= 0.25 \\ \frac{1}{16} \times W &= 0.25 \\ W &= 0.25 \times 16 \\ W &= 4 \text{ kg}\end{aligned}$$

So, the total weight is **4 kg**.

Now, we find 45% of this weight:

$$\begin{aligned}45\% \text{ of } 4 \text{ kg} &= \frac{45}{100} \times 4 \\ &= \frac{180}{100} \\ &= 1.8 \text{ kg}\end{aligned}$$

Therefore, 45% of this weight is **1.8 kg**.

Question 10

In a class of 60 pupils, 15% remained absent on a rainy day. How many pupils were present in the class on that day?

Solution:

Given:

$$\begin{aligned}\text{Total number of pupils} &= 60 \\ \text{Percentage of pupils absent} &= 15\%\end{aligned}$$

The percentage of pupils who were present is:

$$\text{Percentage of pupils present} = 100\% - 15\% = 85\%$$

Now, let us calculate the number of present pupils:

$$\begin{aligned}\text{Number of pupils present} &= 85\% \text{ of } 60 \\ &= \frac{85}{100} \times 60 \\ &= \frac{17}{20} \times 60 \\ &= 17 \times 3 \\ &= 51\end{aligned}$$

Therefore, **51** pupils were present in the class on that day.

Question 11

The monthly income of Mr. Amit Goel is Rs 30400. He saves 12.5% of his income and the rest he spends. How much does he spend each month?

Solution:

Given:

Monthly income of Mr. Amit Goel = Rs 30400

Savings percentage = 12.5%

The spending percentage is:

Spending percentage = $100\% - 12.5\% = 87.5\%$

Now, we calculate the amount spent each month:

Monthly expenditure = 87.5% of Rs 30400

$$\begin{aligned} &= \frac{87.5}{100} \times 30400 \\ &= 87.5 \times 304 \\ &= \text{Rs } 26600 \end{aligned}$$

Alternative Method (by calculating savings first):

Monthly savings = 12.5% of Rs 30400

$$= \frac{12.5}{100} \times 30400 = \frac{1}{8} \times 30400 = \text{Rs } 3800$$

Monthly expenditure = Total income – Savings

$$= 30400 - 3800 = \text{Rs } 26600$$

Therefore, Mr. Amit Goel spends **Rs 26,600** each month.

Question 12

An ore contains 15% iron. How much ore will be required to get 18 kg of iron?

Solution:

Let the quantity of ore required be x kg.

According to the question, 15% of this ore must be equal to 18 kg:

$$15\% \text{ of } x = 18$$

$$\frac{15}{100} \times x = 18$$

$$x = 18 \times \frac{100}{15}$$

$$x = 6 \times \frac{100}{5}$$

$$x = 6 \times 20$$

$$x = 120 \text{ kg}$$

Therefore, **120 kg** of ore will be required to get 18 kg of iron.

Question 13

A property dealer charges a commission of 2% on the first Rs 25000 and 1.5% on the remainder. What commission does he charge for selling a plot of land for Rs 130000?

Solution:

Given:

Total selling price of the land = Rs 130000

Commission on the first Rs 25000 = 2%

Commission on the remaining price = 1.5%

Let us compute the commission in two parts:

1. Commission on the first Rs 25,000:

$$\begin{aligned}\text{Commission}_1 &= 2\% \text{ of } 25000 \\ &= \frac{2}{100} \times 25000 = \text{Rs } 500\end{aligned}$$

2. Commission on the remaining amount: First, find the remaining amount:

$$\text{Remaining amount} = 130000 - 25000 = \text{Rs } 105000$$

Now, calculate the commission on this remainder:

$$\begin{aligned}\text{Commission}_2 &= 1.5\% \text{ of } 105000 \\ &= \frac{1.5}{100} \times 105000 \\ &= 1.5 \times 1050 = \text{Rs } 1575\end{aligned}$$

Total commission charged:

$$\begin{aligned}\text{Total Commission} &= \text{Commission}_1 + \text{Commission}_2 \\ &= 500 + 1575 \\ &= \text{Rs } 2075\end{aligned}$$

Therefore, the property dealer charges a total commission of **Rs 2,075**.

Question 14

In an examination, the maximum marks are 850. Rohit gets 34% marks and fails by 17 marks. Find (i) the passing marks and (ii) the minimum percentage for passing the examination.

Solution:

Given:

Maximum marks = 850

Percentage of marks obtained by Rohit = 34%

First, calculate the actual marks obtained by Rohit:

$$\text{Marks obtained} = 34\% \text{ of } 850 = \frac{34}{100} \times 850 = 34 \times 8.5 = 289 \text{ marks}$$

Since Rohit fails by 17 marks, the passing marks must be 17 more than his score:

(i) **Passing marks:**

$$\text{Passing marks} = 289 + 17 = 306 \text{ marks}$$

(ii) **Minimum passing percentage:**

$$\begin{aligned}\text{Passing percentage} &= \left(\frac{\text{Passing marks}}{\text{Maximum marks}} \times 100 \right) \% \\ &= \left(\frac{306}{850} \times 100 \right) \% \\ &= \left(\frac{3060}{85} \right) \% \\ &= 36\%\end{aligned}$$

Therefore, (i) the passing marks are **306**, and (ii) the minimum percentage for passing the examination is **36%**.

Question 15

A student secures 90%, 60% and 54% marks in three test papers with maximum marks 100, 150 and 200 respectively. Find his aggregate percentage.

Solution:

Let us calculate the marks obtained in each paper:

$$\text{Marks in Paper 1 (out of 100)} = 90\% \text{ of } 100 = \frac{90}{100} \times 100 = 90$$

$$\text{Marks in Paper 2 (out of 150)} = 60\% \text{ of } 150 = \frac{60}{100} \times 150 = 90$$

$$\text{Marks in Paper 3 (out of 200)} = 54\% \text{ of } 200 = \frac{54}{100} \times 200 = 108$$

Now, calculate the total maximum marks and total marks secured:

$$\text{Total maximum marks} = 100 + 150 + 200 = 450$$

$$\text{Total marks secured} = 90 + 90 + 108 = 288$$

Finally, we calculate the aggregate percentage:

$$\begin{aligned}\text{Aggregate percentage} &= \left(\frac{\text{Total marks secured}}{\text{Total maximum marks}} \times 100 \right) \% \\ &= \left(\frac{288}{450} \times 100 \right) \% \\ &= \left(\frac{2880}{45} \right) \% \\ &= 64\%\end{aligned}$$

Therefore, the student's aggregate percentage is **64%**.

Question 16

In an examination, 5% of the applicants were found ineligible and 85% of the eligible candidates belonged to the general category. If 4275 candidates belonged to other categories, then how many candidates applied for the examination?

Solution:

Let the total number of candidates who applied for the examination be x .

1. Since 5% of the applicants were ineligible, the percentage of eligible candidates is:

$$\text{Percentage of eligible candidates} = 100\% - 5\% = 95\%$$

$$\text{Number of eligible candidates} = 95\% \text{ of } x = 0.95x$$

2. Out of these eligible candidates, 85% belong to the general category. Therefore, the percentage of eligible candidates belonging to other categories is:

$$\text{Percentage in other categories} = 100\% - 85\% = 15\%$$

The actual number of eligible candidates in other categories is:

$$\begin{aligned}\text{Number of candidates in other categories} &= 15\% \text{ of (Eligible candidates)} \\ &= 15\% \text{ of } (0.95x) \\ &= \frac{15}{100} \times 0.95x = 0.1425x\end{aligned}$$

We are given that the number of candidates in other categories is 4,275:

$$\begin{aligned}0.1425x &= 4275 \\ x &= \frac{4275}{0.1425} \\ x &= \frac{4275 \times 10000}{1425} \\ x &= 3 \times 10000 \quad (\text{since } 1425 \times 3 = 4275) \\ x &= 30000\end{aligned}$$

Therefore, the total number of candidates who applied for the examination is **30,000**.

Question 17

Gun-powder contains 75% nitre, 10% sulphur and the rest of it is charcoal. Find the quantity of charcoal in 8 kg of gun-powder.

Solution:

Given:

$$\text{Percentage of nitre} = 75\%$$

$$\text{Percentage of sulphur} = 10\%$$

$$\text{Total weight of gun-powder} = 8 \text{ kg}$$

Since the rest of the gun-powder is charcoal, we calculate its percentage:

$$\begin{aligned}\text{Percentage of charcoal} &= 100\% - (\text{Percentage of nitre} + \text{Percentage of sulphur}) \\ &= 100\% - (75\% + 10\%) \\ &= 100\% - 85\% = 15\%\end{aligned}$$

Now, we calculate the actual weight of charcoal in 8 kg of gun-powder:

$$\begin{aligned}\text{Quantity of charcoal} &= 15\% \text{ of } 8 \text{ kg} \\ &= \frac{15}{100} \times 8 \text{ kg} \\ &= \frac{120}{100} \text{ kg} \\ &= 1.2 \text{ kg}\end{aligned}$$

Therefore, the quantity of charcoal in 8 kg of gun-powder is **1.2 kg**.

Question 18

An alloy consists of 13 parts of copper, 7 parts of zinc and 5 parts of nickel. Find the percentage of copper in the alloy.

Solution:

Given the composition of the alloy:

$$\text{Copper} = 13 \text{ parts}$$

$$\text{Zinc} = 7 \text{ parts}$$

$$\text{Nickel} = 5 \text{ parts}$$

$$\text{Total parts in the alloy} = 13 + 7 + 5:$$

$$\text{Total parts} = 25$$

The percentage of copper in the alloy is calculated as:

$$\begin{aligned}\text{Percentage of copper} &= \left(\frac{\text{Parts of copper}}{\text{Total parts}} \times 100 \right) \% \\ &= \left(\frac{13}{25} \times 100 \right) \% \\ &= (13 \times 4) \% \\ &= 52\%\end{aligned}$$

Therefore, the percentage of copper in the alloy is **52%**.

Question 19

Two candidates A and B contested an election. The total votes polled were 9650. If A got 54% of the votes, find the number of votes received by each.

Solution:

Given:

$$\text{Total votes polled} = 9650$$

$$\text{Percentage of votes received by A} = 54\%$$

We calculate the number of votes received by Candidate A:

$$\begin{aligned}\text{Votes received by A} &= 54\% \text{ of } 9650 \\ &= \frac{54}{100} \times 9650 \\ &= \frac{54 \times 965}{10} \\ &= \frac{52110}{10} = 5211 \text{ votes}\end{aligned}$$

Now, the percentage of votes received by Candidate B:

$$\text{Percentage of votes received by B} = 100\% - 54\% = 46\%$$

We calculate the number of votes received by Candidate B:

$$\begin{aligned}\text{Votes received by B} &= 46\% \text{ of } 9650 \\ &= \frac{46}{100} \times 9650 \\ &= \frac{46 \times 965}{10} \\ &= \frac{44390}{10} = 4439 \text{ votes}\end{aligned}$$

Verification:

$$\text{Total votes} = \text{Votes for A} + \text{Votes for B} = 5211 + 4439 = 9650$$

Therefore, Candidate A received **5211 votes** and Candidate B received **4439 votes**.

Question 20

At an election between two candidates, 68 votes were declared invalid. The winning candidate secures 52% of the valid votes and wins by 354 votes. Find the total number of votes polled.

Solution:

Let the total number of valid votes be V .

1. Percentage of valid votes received by the winning candidate = 52%.
2. Since there are only two candidates, the percentage of valid votes received by the losing candidate:

$$\text{Percentage of votes for loser} = 100\% - 52\% = 48\%$$

3. The winner wins by a margin of:

$$\text{Margin percentage} = 52\% - 48\% = 4\% \text{ of valid votes}$$

According to the question, the winner wins by 354 votes:

$$4\% \text{ of } V = 354$$

$$\frac{4}{100} \times V = 354$$

$$V = 354 \times \frac{100}{4}$$

$$V = 354 \times 25$$

$$V = 8850 \text{ valid votes}$$

The total number of votes polled is the sum of valid votes and invalid votes:

$$\text{Total votes polled} = \text{Valid votes} + \text{Invalid votes}$$

$$= 8850 + 68$$

$$= 8918 \text{ votes}$$

Therefore, the total number of votes polled is **8918**.

Question 21

The salary of Mrs Sarita is Rs 32000 per month. 10% of it is deducted by the employer as provident fund. Of the remaining money, she spends 20% on house rent, 46% on food, 14% on the education of children and 10% on other expenses. Rest she saves. Find:

- (i) how much is credited each month to her Provident Fund Account.
- (ii) how much is spent on food.
- (iii) how much is paid as house rent.
- (iv) how much is spent on the education of children.
- (v) how much does she save every month.

Solution:

Given:

Mrs Sarita's monthly salary = Rs 32000

PF deduction rate = 10% of total salary

Let us answer each part step-by-step:

(i) Provident Fund deduction:

$$\text{PF deduction} = 10\% \text{ of Rs } 32000 = \frac{10}{100} \times 32000 = \text{Rs } 3200$$

So, **Rs 3,200** is credited each month to her Provident Fund Account.

(ii) Calculate the remaining money after PF deduction:

$$\text{Remaining money} = \text{Total salary} - \text{PF deduction} = 32000 - 3200 = \text{Rs } 28800$$

All subsequent expenditures are calculated as percentages of this **remaining money** (Rs 28,800).

Amount spent on food (46%):

$$\text{Spent on food} = 46\% \text{ of Rs } 28800 = \frac{46}{100} \times 28800 = 46 \times 288 = \text{Rs } 13248$$

So, she spends **Rs 13,248** on food.

(iii) Amount paid as house rent (20%):

$$\text{Spent on house rent} = 20\% \text{ of Rs } 28800 = \frac{20}{100} \times 28800 = 20 \times 288 = \text{Rs } 5760$$

So, she pays **Rs 5,760** as house rent.

(iv) Amount spent on the education of children (14%):

$$\text{Spent on education} = 14\% \text{ of Rs } 28800 = \frac{14}{100} \times 28800 = 14 \times 288 = \text{Rs } 4032$$

So, she spends **Rs 4,032** on the education of children.

- (v) **Monthly savings:** First, find the total percentage of the remaining money spent on various items:

$$\begin{aligned}\text{Total spent percentage} &= 20\%(\text{rent}) + 46\%(\text{food}) + 14\%(\text{education}) + 10\%(\text{others}) \\ &= 90\%\end{aligned}$$

Thus, her savings percentage of the remaining money is:

$$\text{Savings percentage} = 100\% - 90\% = 10\%$$

Now, calculate the actual savings:

$$\text{Monthly savings} = 10\% \text{ of Rs } 28800 = \frac{10}{100} \times 28800 = \text{Rs } 2880$$

So, she saves **Rs 2,880** every month.
